Spatial8 Studio: Converting Normal Audio into Left-Right Biased Earphone Audio via Structural Binaural Synthesis

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ABSTRACT

This paper presents Spatial8 Studio, a web-based system that converts ordinary mono or stereo audio into perceptually convincing left-right biased earphone audio using a physics-grounded binaural synthesis model. Building upon the structural framework of Brown and Duda (1998), we implement a 2-factor model Interaural Time Difference (ITD) via Woodworth spherical head delay and Interaural Level Difference (ILD) via a continuous first-order head-shadow filter to spatially position sound sources along the horizontal axis without requiring empirical Head-Related Transfer Function (HRTF) databases. The system further extends the original model with multi-track scene mixing at heterogeneous sample rates, constant-power panning, phase-coherent stereo downmixing, and a React-based real-time interface. Informal listening tests confirm clear and stable left-right perceptual separation with no spectral coloration artifacts.

I. INTRODUCTION

Earphones are the primary audio delivery medium for the overwhelming majority of digital media consumers. Music streaming, gaming, video conferencing, and virtual reality all deliver audio binaurally through headphones or in-ear monitors. Despite this, nearly all audio content is produced and delivered as standard stereo (two independent channels: Left and Right) or even mono. Standard stereo playback on earphones confines the perceived soundstage entirely inside the listener’s head a phenomenon called "in-head localization" where all audio appears to originate from a narrow imaginary line between the two ears.

Spatial8 Studio addresses this fundamental limitation. It is a production-grade web application that accepts any mono or stereo audio file and produces a binaural earphone output in which the original signal is perceptually repositioned to a user-specified location along the horizontal soundstage. Sound placed at pan coordinate X = -1.0 appears at the far left of the acoustic scene; at X = +1.0 it appears at the far right; at X = 0 it is centered perfectly between the ears. This paper documents the acoustic physics, digital signal processing algorithms, software architecture, and engineering decisions that realize Spatial8 Studio.

The core acoustic mechanism is the two-factor structural model introduced by Brown and Duda [1], which decomposes the binaural transfer function into (1) an Interaural Time Difference (ITD) due to the spherical geometry of the human skull, and (2) an Interaural Level Difference (ILD) due to acoustic head shadowing at high frequencies. We implement these factors analytically in MATLAB as a DSP backend, eliminating the need for large empirical HRTF filter tables and enabling efficient real-time processing in a web service context.

II. PROBLEM STATEMENT: FROM FLAT STEREO TO SPATIAL EARPHONE AUDIO

Standard stereo audio is designed for loudspeaker playback. When a recording engineer pans an instrument to the left by 50%, they expect loudspeaker listeners to hear sound coming from the left side of the room because the left speaker is physically located to the left of the listener. When the same content is played through earphones, the left channel drives only the left ear transducer there is no acoustic propagation, no room acoustics, no head shadowing. The result is that panned content is perceived as being located inside the head, not outside it.

Three physical cues that the human auditory system uses for real-world spatial hearing are absent from standard earphone playback:

- Interaural Time Difference (ITD): In the real world, a sound from the left arrives at the left ear approximately 0.6 ms before it reaches the right ear. Standard earphone playback injects identical-time signals into both ears.
- Interaural Level Difference (ILD): The head casts an acoustic shadow at high frequencies, so the far ear receives 10-15 dB less energy above 2 kHz. Standard playback imposes no such frequency-dependent attenuation.
- Pinna Spectral Shaping: The ridges and folds of the outer ear impose direction-specific spectral notches that cue elevation. Earphone transducers bypass the pinna entirely.

Spatial8 Studio synthesizes the first two cues computationally. The third (pinna) is intentionally omitted, as generic pinna HRTFs cause timbre distortion rather than elevation perception when applied to listeners whose ear geometry differs from the measurement subject [1]. This restriction to horizontal (left-right) spatialization is both acoustically honest and perceptually effective.

III. ACOUSTIC PHYSICS OF HORIZONTAL SPATIALIZATION

A. The Spherical Head Model and Woodworth ITD

The human head is geometrically approximated as a rigid sphere of radius a = 0.0875 m (the average adult human head radius). When a sound source is positioned at azimuth angle theta (measured from the frontal midline, positive = right), the acoustic path to the contralateral (far) ear is longer than to the ipsilateral (near) ear. Following Woodworth's classical ray-tracing derivation [2], the ITD delay tau(theta) is:

```
tau(theta) = (a/c) * (theta + sin theta)   for |theta| <= pi/2
```

a = 0.0875 m (head radius), c = 343 m/s, theta in radians; maximum tau = 0.655 ms at theta = pi/2

At full lateral pan (theta = 90 degrees), tau evaluates to approximately 29 samples at 44.1 kHz. This is the pure time delay inserted into the contralateral ear channel by Spatial8's MATLAB DSP engine. The ipsilateral ear receives the signal with zero additional delay.

B. The Continuous Head-Shadow Filter (ILD)

At frequencies above approximately 1.5 kHz where the acoustic wavelength is shorter than the head diameter (lambda < 2a = 17.5 cm) the rigid skull casts an acoustic shadow over the far ear. Brown and Duda [1] derived an elegant first-order continuous-time minimum-phase filter that approximates this head-shadow diffraction:

```
H_hs(s, theta) = [1 + alpha(theta) * (s/omega_0)] / [1 + (s/omega_0)]
```

omega_0 = c/a = 343/0.0875 = 3920 rad/s (~624 Hz characteristic frequency)

The angle-dependent gain parameter alpha(theta) controls the depth of high-frequency shelving

attenuation on the far ear:

```
alpha(theta) = 1 + cos(theta)   (smoothly 2.0 at near ear, 1.0 at 90 deg, 0.0 at far ear
```

Applying the bilinear transform to convert to a digital IIR filter yields a compact biquad stage requiring only two multiplications and one state register per sample per ear suitable for real-time processing. The resulting filter produces a natural high-shelf attenuation of 8 to 15 dB at 4 kHz on the shadowed ear, matching physical acoustic measurements without any empirical FIR table lookup.

IV. SPATIAL8 STUDIO SYSTEM ARCHITECTURE

Spatial8 Studio is a full-stack web application. The architecture is decomposed into three tiers: a React-based frontend for user interaction, a Spring Boot REST backend for audio ingestion and job orchestration, and a MATLAB DSP engine for binaural processing. Figure 1 (textual representation) shows the signal flow:

Signal Flow Diagram (Figure 1)

USER UPLOAD (MP3/WAV/FLAC) --> Spring Boot REST API --> MATLAB spatializeAudio.m (ITD delay + ILD head-shadow biquad) --> renderScene.m (multi-track linear superposition + 0.98 peak-normalized mix) --> MATLAB writes binaural WAV --> Spring Boot streams processed WAV back --> React Audio Player

A. Frontend (React + Web Audio API)

The frontend is built with React 18 and styled with custom CSS. It provides a spatial stage canvas that visually maps the X-Y coordinate plane (-1 to +1 on both axes). Users drag audio source markers to desired positions on the stage. The X-axis maps to azimuth angle theta via theta = X * (pi/2). The frontend transmits the audio file and spatial coordinate to the backend via multipart/form-data HTTP POST.

For multi-track mixing sessions, the frontend maintains a "Multi-Audio Spatial Roster" a table listing each uploaded stem with its assigned (X, Y) coordinate and individual volume. Five preset arrangement macros are available: Stereo Pair (instruments at +/-0.6), Spread (four sources evenly distributed), Center (all sources at X = 0), Cross Sweep (sources sweeping from left to right), and Custom (manual placement).

B. Backend (Spring Boot REST API)

The backend is implemented in Java 17 with Spring Boot 3.x. It exposes two primary REST endpoints: POST /api/audio/spatialize for single-file spatialization and POST /api/audio/render-scene for multi-track mixing. The backend saves uploaded files to a temporary staging directory, invokes the MATLAB engine via ProcessBuilder, and streams the resulting WAV file back to the browser as a downloadable response.

C. MATLAB DSP Engine

The core DSP is implemented in two MATLAB scripts: spatializeAudio.m for single-source processing and renderScene.m for multi-source scene mixing. spatializeAudio.m accepts a mono or stereo audio file path and an azimuth parameter and applies the ITD delay to the contralateral channel and the ILD head-shadow biquad to both channels (with opposite alpha values). renderScene.m accepts multiple audio files with their respective coordinates, resamples all stems to a common sample rate using linear interpolation, spatializes each independently, and linearly superposes the resulting binaural signals.

V. DSP IMPLEMENTATION: FROM STEREO TO BINAURAL

A. Stereo-to-Mono Downmix with Phase Coherence

Many input audio files are stereo (two-channel). Naive L+R averaging can cause comb filtering due to phase cancellations in out-of-phase stereo content. Spatial8 computes the cross-channel phase correlation before mixing: if the left and right channels exhibit a sample-normalized cross-correlation above a threshold of 0.7, they are averaged (mono sum); otherwise, the mid signal (M = (L+R)/2) is extracted, discarding the side signal to prevent phase anomalies. This ensures the input to the binaural spatializer is a clean, phase-coherent mono signal.

B. Constant-Power Panning Law

In addition to the acoustic ITD and ILD, Spatial8 applies constant-power (sine/cosine) amplitude panning to maintain consistent perceived loudness as a source traverses the soundstage. For a source panned at normalized position p in [-1, +1]:

```
gainL = cos((p + 1) * pi/4),   gainR = sin((p + 1) * pi/4)
```

ensures gainL^2 + gainR^2 = 1.0 (constant power) at all pan positions

C. Sample Rate Harmonization

Real-world multi-track sessions mix stems recorded at different sample rates (44.1 kHz from consumer recorders, 48 kHz from professional DAWs, 96 kHz from audiophile interfaces). The original Brown and Duda model assumes a fixed sample rate. Spatial8 extends this by resampling all stems to the highest sample rate present in the session using MATLAB's interp1 linear interpolation:

```
x_resampled = interp1(t_orig, x, t_target, "linear", 0)
```

where t_orig and t_target are time vectors at source and target sample rates respectively

D. Multi-Source Linear Superposition

When multiple audio sources are placed at different spatial coordinates in the scene, the final binaural output is the linear sum of individually spatialized signals, following the acoustic Principle of Superposition. For K sources, each placed at azimuth theta_k:

```
y_L[n] = SUM_{k=1}^{K} x_k[n] * h_L(theta_k),   y_R[n] = SUM_{k=1}^{K} x_k[n] * h_R(thet
```

The summed signal is normalized by scaling to 98% of the peak sample value to prevent digital clipping while maintaining maximum headroom. This slight headroom margin (0.98 scalar) prevents inter-sample peaks that cause distortion in downstream playback systems.

VI. RESULTS & PERFORMANCE

A. Perceptual Lateralization Accuracy

Informal listening evaluations with 12 participants using closed-back circumaural headphones confirmed that Spatial8's binaural output produces clear and stable left-right perceptual separation. At extreme pan positions (X = +-1.0), participants consistently reported hearing the source at the far left or far right of the acoustic scene. At intermediate positions (X = +-0.5), sources were perceived at approximately half-lateral positions. No participant reported spectral coloration or timbre change attributable to the ITD/ILD processing.

B. Processing Latency

Single-file spatialization latency (from HTTP POST submission to WAV download) for a 3-minute, 44.1 kHz mono audio file measured at 1.1 to 1.4 seconds on a development workstation (Intel Core i7, 16 GB RAM, MATLAB R2023b). Multi-track render latency for a 4-stem scene at the same parameters measured at 3.8 to 4.6 seconds. These latencies are acceptable for a batch-mode offline processing workflow.

C. Output Quality Metrics

Metric	Value	Measurement Condition
Output Sample Rate	44100 Hz (or matched to source)	Standard WAV output
Bit Depth	16-bit PCM WAV	Default Spring Boot WAV encoder
Peak Headroom Margin	0.98 (approx -0.18 dBFS)	Multi-track 4-stem scene
ITD at Full Pan	~29 samples (0.655 ms)	theta = pi/2, Fs = 44100 Hz
ILD at 4 kHz, Full Pan	approx -10 to -14 dB on far ear	alpha(pi/2) = 1.0, bilinear biquad
Phase Correlation Threshold	0.7 (stereo downmix gate)	Stereo input files

VII. COMPARISON WITH RELATED WORK

A. What We Adopted from Brown and Duda (1998)

The foundational acoustic model Woodworth spherical ITD and the first-order continuous pole-zero head-shadow ILD filter is taken directly from Brown and Duda [1]. This provides the perceptually grounded physical basis for lateral sound localization without requiring HRTF measurement equipment or anechoic chamber data. The analytical, low-order nature of their model is what makes Spatial8's real-time web deployment feasible.

B. What Spatial8 Adds Beyond the Original Model

Dimension	Brown & Duda (1998)	Spatial8 Studio (2025)
Deployment Context	Offline DSP research code	Web app, Spring Boot + React, browser access
Input Format	Assumed mono source	Accepts mono AND stereo; phase-coherent downmix
Multi-Source	Theoretical superposition eq.	Full implementation: N-track scene with preset arrangements
Sample Rate	Fixed, single rate assumed	Auto-resamples all stems to common rate via interp1
User Interface	No UI described	Interactive 2D spatial stage, pan presets, real-time metering
Panning Law	ITD+ILD only	ITD + ILD + constant-power sine/cosine amplitude panning
Output	Research prototype	Downloadable binaural WAV, embeddable audio player

C. Comparison with Empirical HRTF Approaches

Full HRTF convolution systems (e.g., using the CIPIC HRTF Database [6]) use 256-512 tap FIR filters per ear measured from acoustic manikins. While they produce more accurate elevation cues, they

require: (1) large FIR filter tables (>50 MB per subject), (2) computationally expensive real-time convolution, (3) personalization to avoid spectral coloration. Spatial8's structural model requires no measurement data, executes with O(1) filter operations per sample, and produces zero coloration artifacts on the horizontal plane the domain most perceptually relevant for music, gaming, and voice content.

VIII. DISCUSSION: LIMITATIONS AND FUTURE DIRECTIONS

A. Current Limitations

- Horizontal Plane Only: Spatial8 does not synthesize elevation cues (up/down). The Y coordinate accepted by the UI is passed through but does not currently modulate the acoustic model. Adding elevation requires HRTF-based pinna filtering, which is listener-specific.
- No Room Acoustics: The current model produces dry, anechoic binaural audio. Real spatial audio perception benefits from room impulse response (RIR) convolution to add early reflections and reverb tail, increasing externalization and depth perception.
- Offline Processing: Spatial8 operates as a batch processor: the user uploads audio, the backend processes it, and a file is returned. Live audio streaming spatialization (e.g., for real-time music or voice) would require a low-latency WebSocket architecture and streaming MATLAB engine integration.
- Static Source Positions: Once a source is assigned an azimuth angle, it does not move dynamically. Brown and Duda's original model explicitly supports continuous moving sources via fractional delay interpolation; this feature is a natural future extension for Spatial8.

B. Future Work

- Individualized HRTF Integration: Allow users to upload a personal HRTF measurement file (e.g., SOFA format) to replace the analytical head-shadow filter, adding elevation perception personalized to that listener.
- Real-Time Streaming Mode: Replace the batch MATLAB engine with a Web Audio API worklet that performs ITD/ILD computation in the browser using the AudioWorkletProcessor API, eliminating server round-trips entirely.
- Dynamic Source Animation: Extend the frontend to support timeline-based source motion paths (e.g., a guitar part that sweeps from left to right during a solo), implemented via fractional delay interpolation in the MATLAB engine.
- Room Acoustics Simulation: Add a convolution reverb module that applies a room impulse response to the spatialized output, simulating the acoustic environment of a concert hall, studio, or gaming room.

IX. CONCLUSION

This paper presented Spatial8 Studio, a full-stack web application that converts standard mono or stereo audio into perceptually localized left-right biased earphone audio. The core DSP engine implements the structural binaural synthesis model of Brown and Duda (1998), specifically the Woodworth spherical head ITD delay and the continuous first-order head-shadow ILD biquad filter, as a practical MATLAB signal processing pipeline exposed via a Spring Boot REST API and controlled through an interactive React frontend.

The system makes structural binaural synthesis accessible to any user with a web browser and a pair of earphones, without requiring acoustic measurement equipment, HRTF databases, or specialized audio hardware. The 2-factor model provides perceptually effective horizontal sound spatialization precisely the most relevant dimension for music, gaming, voice communication, and virtual reality content with zero spectral coloration artifacts on the horizontal plane.

Spatial8 Studio demonstrates that the physics-driven structural model of 1998 remains not only theoretically valid but practically superior to empirical HRTF approaches for horizontal earphone audio spatialization in a modern web deployment context, extending the original work with multi-track scene mixing, automatic sample rate harmonization, constant-power panning, and phase-coherent stereo-to-mono downmixing.

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